

## Programming and Problem Solving through Python Language O Level / A Level

### Chapter -1 : Introduction to Programming

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#### Algorithm

A sequential solution of any program that written in human language, called algorithm.

Algorithm is first step of the solution process, after the analysis of problem, programmer writes the algorithm of that problem.

#### 3. Write an algorithm to find all roots of a quadratic equation $ax^2+bx+c=0$ .

Step 1: Start

Step 2: Declare variables a, b, c, D, r1, r2, rp and ip;

Step 3: Calculate discriminant

$$D \leftarrow b^2 - 4ac$$

Step 4: If  $D \geq 0$

$$r1 \leftarrow (-b + \sqrt{D}) / 2a$$

$$r2 \leftarrow (-b - \sqrt{D}) / 2a$$

Display r1 and r2 as roots.

Else

Calculate real part and imaginary part

$$rp \leftarrow b / 2a$$

$$ip \leftarrow \sqrt{(-D)} / 2a$$

Display  $rp + j(ip)$  and  $rp - j(ip)$  as roots

Step 5: Stop

#### 4. Write an algorithm to find the factorial of a number entered by the user.

Step 1: Start

Step 2: Declare variables n, factorial and i.

Step 3: Initialize variables

factorial  $\leftarrow$  1

i  $\leftarrow$  1

Step 4: Read value of n

Step 5: Repeat the steps until i = n

5.1: factorial  $\leftarrow$  factorial\*i

5.2: i  $\leftarrow$  i+1

Step 6: Display factorial

Step 7: Stop

#### Factorial of Number

$$5 = 1 \times 2 \times 3 \times 4 \times 5 = 120$$

$$7 = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7$$

**5. Write an algorithm to check whether a number entered by the user is prime or not.**

Step 1: Start

Step 2: Declare variables n, i, flag.

Step 3: Initialize variables

flag  $\leftarrow$  1

i  $\leftarrow$  2

Step 4: Read n from user.

Step 5: Repeat the steps until  $i < (n/2)$

5.1 If remainder of  $n \div i$  equals 0

flag  $\leftarrow$  0

Go to step 6

5.2 i  $\leftarrow$  i+1

Step 6: If flag = 0

Display n is not prime

else

Display n is prime

Step 7: Stop

## **6. Write an algorithm to find the Fibonacci series till term $\leq 1000$ .**

Step 1: Start

Step 2: Declare variables first\_term, second\_term and temp.

Step 3: Initialize variables first\_term  $\leftarrow 0$  second\_term  $\leftarrow 1$

Step 4: Display first\_term and second\_term

Step 5: Repeat the steps until second\_term  $\leq 1000$

5.1: temp  $\leftarrow$  second\_term

5.2: second\_term  $\leftarrow$  second\_term + first\_term

5.3: first\_term  $\leftarrow$  temp

5.4: Display second\_term

Step 6: Stop

### **Assignment**

**Q.1 Write an algorithm for printing first 10 natural number.**

**Q.2 Write an algorithm for printing first 10 odd number.**

**Q.3 Write an algorithm for printing first 10 even number.**

**Q.4 Write an algorithm for printing first 10 natural number in reverse order.**